BlenderMesh2STEP

Turn Blender meshes into real CAD STEP solids

User Guide — version 0.11.0

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Abstract

BlenderMesh2STEP is a Blender 4.2+ extension that converts triangle meshes into genuine STEP AP242 B-rep solids — the kind FreeCAD, Fusion 360, SolidWorks and Onshape open as editable, measurable, manufacturable geometry. You select the faces of a feature, the extension fits the exact analytic surface (plane, box, cylinder, cone, sphere, torus, fillet) by least squares — typically to machine precision on clean Blender-authored meshes — and the exporter writes a true solid, with real holes and pockets cut by boolean operations. This guide covers installation, the reverse (mesh $\rightarrow$ CAD) workflow, forward parametric modeling, exporting, and verifying the result.

Contents

1 Introduction

1.1 The problem

A mesh has thrown away the *intent* you want back: a 20 mm cylinder and a 64-sided prism are identical triangle soups. Exporting STL (or tessellated STEP) hands a machine shop a faceted blob they cannot measure, fillet, or edit.

BlenderMesh2STEP does not guess — **you point, it reconstructs**. You tell it "these faces are a cylinder," and it recovers the exact analytic surface by least-squares fitting. The result is a true boundary-representation (B-rep) solid with real `CYLINDRICAL_SURFACE` entities, not an approximation. On clean meshes the fits land at $\sim 10^{-8}$ RMS — effectively exact.

1.2 Two workflows

Reverse (mesh → CAD)
: Select the faces of one feature in Edit Mode, fit a primitive, repeat per feature, export. This is the main workflow and the subject of Section 4.

Forward (build → CAD)
: Add parametric STEP-primitive solids directly from the **Build — STEP Primitives** panel. These are STEP-exact *by construction*; the viewport mesh is only a preview. See Section 6.

Both workflows feed the same *feature stack* and the same exporter, and can be mixed freely — e.g. fit a base plate from a mesh, then add a forward-built cylinder as a *Subtract* feature to drill a hole.

1.3 Two export backends

OCCT kernel (recommended)
: Uses OpenCASCADE — the kernel FreeCAD is built on — installed on demand from the panel (Section 2.3). This is the path that can *cut holes* (booleans), merge bodies into one watertight solid, heal gaps, and validate the result (per-solid volume and topology checks).

Pure-Python writer (fallback)
: Zero dependencies. Writes real analytic AP242 solids, but has **no boolean kernel**: subtractive features cannot be cut from the part (Section 7.6).

> **Important**
>
> If your part has holes, pockets, counterbores, or must be a single watertight body, install the OCCT kernel. Without it a "hole" can only be marked, skipped, or written as filled material — never actually cut.

2 Installation

2.1 Requirements

- Blender **4.2 or newer** (the extension uses the 4.2+ extension format).
- No other dependencies for fitting and the pure-Python STEP writer.
- Optionally, the OCCT kernel ($\sim$100 MB, one click) for booleans, watertight sewing, and validation.

2.2 Installing the extension

1. Download `reverse_mesh-0.11.0.zip` from the project's GitHub Releases page.
2. In Blender: **Edit** → **Preferences** → **Get Extensions** → (top-right dropdown) **Install from Disk...** and pick the zip.
3. Enable it if it is not enabled automatically. The **Reverse** tab appears in the 3D-viewport sidebar (press `N`).

2.3 Installing the OCCT kernel

In the **Reverse** tab, open **Fitted Features** → **Export** and click **Install OCCT**. This downloads the `cadquery-ocp` binding into the extension's own directory (it does not touch your system Python). When it is ready the panel shows **OCCT ready**. All boolean and watertightness options in the export dialog are marked *(OCCT only)* and silently do nothing on the pure-Python path.

3 Quick start: the 60-second workflow

1. Select your mesh object, enter **Edit Mode**, open the **Reverse** tab in the sidebar (`N`).
2. Choose a **Primitive** (or leave **Auto-detect**) and a **Role**: *Add* for material, *Subtract* for a hole/pocket cutter.
3. Select the faces of *one* feature. Click one face of a wall and use **Select Similar** to grow the selection across the whole surface (it stops at sharp edges).
4. Click **Fit Primitive to Selection**. The fit lands in the **Fitted Features** stack with its RMS error; the optional heatmap colours the faces by deviation. Repeat for each feature.
5. Click **Export STEP (AP242)**, review the options (Section 7), export, and read the **Validation Report** panel to confirm volumes and watertightness.

4 The reverse workflow in detail

4.1 Primitives you can fit

Primitive	Recovers	Notes
Plane	flat face with extent	exported as a bounded planar patch
Box	oriented cuboid (rotation too)	fit from several flat faces at once
Cylinder	axis, radius, height	the workhorse for holes and bosses
Cone	apex, half-angle, radii	full cones and frustums
Sphere	centre, radius	
Torus	axis, major + minor radius	best on full rings
Fillet	edge round → partial cylinder	radius + arc extent; arcs < 330°
Extrude	prism: planar line/arc profile + height	select the whole prism incl. caps
Revolve	lathe part: line/arc profile × 360°	explicit only; never chosen by Auto
Auto	best of the above	normal-agreement gate + simplicity tie-break

Table 1: Fittable primitives. All fits report RMS and maximum error.

The **Extrude** fit recovers the archetypal Blender-authored mechanical part: a planar outline of lines and arcs swept along an axis (brackets, plates, slots, bosses). Select the *entire* prism — side walls and both end caps — and the fitter recovers the axis, the height, and an exact line/arc profile (tangent arcs such as slot ends are recognised as true arcs, not chains of facets). It exports as a genuine B-rep solid on both backends and can act as a *Subtract* cutter (a milled pocket).

The **Revolve** fit recovers lathe parts: select the whole revolved body and the fitter solves the rotation axis directly from the surface normals, then reconstructs the (radius, height) profile — lines and arcs, closed along the axis when the solid touches it. Washers, bushings, pulleys and ball-end pins export as exact solids of revolution on both backends (cylindrical, conical, planar, toroidal and spherical zones), and a subtractive revolve cuts grooves. Because *every* quadric is a surface of revolution, Revolve is never chosen by **Auto-detect** — pick it explicitly.

Auto-detect tries every primitive and keeps the best fit, preferring the *simplest* primitive when several explain the selection equally well (a flat patch becomes a plane, not a giant sphere). The runner-up candidates are shown for the active feature so you can see how close the call was.

4.2 Selecting a feature's faces

- **Select Similar** grows the selection from the active face across edges whose face-normal angle is below the *crease angle* — one click on a cylindrical wall selects the whole wall and stops at the caps.
- **Segment regions** (in the main panel) splits a multi-surface selection into smooth-connected regions and fits each separately — e.g. a whole cube selection becomes 6 planes; a full cylinder becomes side + 2 caps.

4.3 Fit quality controls

Fit-quality heatmap
: Colours the selected faces green → red by deviation from the fitted surface, so a stray chamfer or mis-picked face is obvious instead of buried in an average.

Reject outliers (RANSAC)
: Trims faces that deviate from the consensus surface and refits on the rest. Use when a selection is slightly dirty (an odd triangle, the start of a chamfer).

Snap dimensions
: Rounds fitted radii/lengths to a nice value (e.g. 19.98 → 20.0) when they are already very close, so the STEP carries manufacturable numbers. Presets 0.1 / 0.5 / 1.0 or a custom step.

Tolerance
: The RMS threshold (as a fraction of region size) above which the fit is flagged as poor.

4.4 Propagate Pattern: fit one hole, get the rest

Fit *one* hole (a subtractive cylinder), then click **Propagate Pattern**. The extension scans the mesh for every matching hole — same radius and axis direction within tolerance — and fits them all with the seed's settings. Bolt circles, linear arrays and mirrored holes are recognised and labelled (circular / linear / scattered).

4.5 Whole-mesh auto-decompose (experimental)

Two heavier, whole-mesh paths exist for when you want a first pass without hand-selecting anything. Both are marked red in the UI because they optimise over the *entire* mesh and can take a while:

Auto-Decompose Whole Mesh
: Surface decomposition: sweeps coarse→fine crease angles, builds competing primitive hypotheses, and selects a set by global energy minimisation (data fit + coverage + a per-primitive complexity cost + boundary smoothness). Faces no primitive explains can

be kept as a `Reverse_Leftover` faceted patch so the exported STEP still contains the complete part.

Auto-Decompose Solid (Boolean)
Volumetric decomposition: samples the interior volume and greedily covers it with inscribed spheres, cylinders, cones, tori and boxes, producing an additive union of solids. Export with the OCCT **Merge into one solid** option.

Treat both as a convenience to correct, not a promise: review the resulting feature set, delete bad members, and re-fit where needed.

5 The feature stack

Every fit and every forward-built primitive lands in **Fitted Features**. Each entry shows its boolean role (`+`/`-`), kind, a summary with dimensions, and its RMS (or "exact" for built primitives). Prefixes mark machine-generated sets: `[A]` surface auto-decompose, `[S]` solid decompose, `[B]` forward-built.

From the stack you can, non-destructively:

- **Reorder** features (arrows). Order matters with *Booleans in stack order* at export: an Add placed after a cut refills it (e.g. a boss inside a pocket).
- **Re-fit** a feature from its remembered source faces after editing the mesh.
- **Toggle Add / Subtract**, and for subtractive features choose **Through** (cutter overshoots both ends so the hole opens cleanly) or **Blind** (keeps the pocket depth exact; the open end is auto-detected).
- **Tag threads**: on a cylinder/cone, enter e.g. `M8x1.25`. The designation is written into the STEP product name and the PMI sidecar.
- **Hole presets**: on a subtractive cylinder choose **Counterbore** (recess radius + depth) or **Countersink** (recess radius + angle); the recess is cut as an extra coaxial cutter on the OCCT path.
- **Select** the feature's generated object in the viewport; **delete** single features or **clear** whole sets.

The stack persists in the `.blend` file; the authoritative parameters of each feature live on its generated object, so features survive reloads and renames of the session list.

6 Forward modeling: Build — STEP Primitives

The **Build** panel adds solids that are STEP-exportable by construction: **Box, Cylinder, Cone (frustum), Sphere, Torus, Extrude (N-gon prism), Ring (washer)**. The exact analytic parameters are stored on the object; the viewport mesh is just a preview tessellation. For the extruded prism, pick the side count when adding; the circumradius and height stay live-editable and the profile follows. The ring is a rectangular section revolved about the axis (washers, spacers, bushings) with live inner/outer radius and height.

- Pick a primitive and a **Role** (Add or Subtract — a forward-built cylinder makes a perfect drill), then **Add Primitive**.
- With the object active, its dimensions appear as live-editable fields; editing one regenerates the mesh immediately.
- If you move/rotate the object, that transform is carried into the export. If you *scale* it in Object Mode, the panel offers **Bake Scale into Parameters** — click it so the stored

dimensions match what you see. Non-uniform scale on curved primitives is flagged: it is not representable as an analytic STEP surface.
- If the mesh drifts out of sync with the stored parameters (e.g. you edited vertices by hand), the panel warns and offers **Rebuild from Parameters**.

7 Exporting STEP

Click **Export STEP (AP242)** in the **Fitted Features** panel. The file dialog carries the export options:

7.1 Backend and boolean options

Option	Meaning
Backend	**Auto** (OCCT if installed, else pure Python), or force either.
Merge into one solid	Fuse all Add solids into a single body (OCCT).
Booleans in stack order	Apply Add/Subtract in feature-stack order so a later Add refills a cut. Off = fuse all Adds first, then cut (OCCT).
Fillets as real blends	Apply recognized edge fillets as true rounds on the solid at the fitted radius — the body stays watertight and the blend is editable in CAD. Applies when the export produces a single body (Merge or cutters); fillets whose edge cannot be located fall back to trimmed patches (OCCT).
Cutter overshoot	Extend cutters by this fraction at each end so through holes open cleanly on coplanar faces (OCCT).
Auto-stitch shared edges	Unify coincident faces so abutting features share real edges (OCCT).
Make watertight + Sew tolerance	Sew all faces, build a closed solid, heal gaps up to the tolerance; reports remaining open edges (OCCT).
Cutters (pure Python)	What to do with Subtract features when there is no kernel: **Include marked** (red `cutter:` reference solids — default), **Skip**, or **Include as solids** (holes appear filled).

Table 2: Backend/boolean export options.

7.2 Units and scale

The **Unit** option sets the length unit declared inside the STEP file (mm / m / inch); it defaults to your scene's display unit. **Scale from Scene units** derives the coordinate scale from Blender's unit settings — e.g. a metric scene at Unit Scale 0.001 with STEP unit mm gives `1 BU` → `1 mm`. The dialog always shows the *effective scale* live; check it before exporting. Choose **Manual** to force a factor. Scenes with unit system *None* pass coordinates through unchanged.

7.3 Scope and extras

- **Selected only** exports just the selected Reverse objects.
- **Feature set** filters by group: All / Manual / Built / Auto / Solid — useful when an auto-decompose set coexists with hand fits.
- **Include leftover patches** writes the unexplained-face patches so the STEP contains the complete part (as faceted planar faces).
- **Write PMI sidecar** adds `.pmi.json` and `.pmi.csv` next to the STEP with every feature's dimensions (radii, diameters, lengths, angles, threads) and pairwise relationships (axis angles, hole spacing) — handy for inspection sheets.

- **Embed semantic PMI** embeds AP242 `DIMENSIONAL_SIZE` entities so CAD reads diameters/lengths as queryable PMI (pure-Python writer only).

7.4 Named bodies and colours

On the OCCT path, exports are written through the XCAF document layer: every body becomes its *own named product* (the feature or object name) and per-feature colours are embedded so they survive into FreeCAD/Fusion. With **Merge into one solid** the merged body takes the file's name. The **Make watertight** pass rebuilds geometry as one healed body and therefore falls back to a single unnamed product.

7.5 The validation report

After an OCCT export, the **Validation Report** sub-panel lists every solid with its volume and validity, plus a watertightness line (open-edge count if not). Read it every time: *valid + watertight + plausible volume* is your evidence the file will survive a CAD import and a machine shop's quote. The pure-Python path cannot validate; it says so instead of pretending.

7.6 Limits of the pure-Python writer

The built-in writer emits real analytic solids (box, cylinder, cone, sphere, torus), bounded plane patches, trimmed partial-cylinder fillet patches, and faceted leftover patches — with per-feature colours. It cannot: perform booleans (no holes), merge bodies, sew to watertight, or validate. Overlapping Add bodies are written as separate interpenetrating solids in one part.

8 Verification and troubleshooting

8.1 Inspecting a part in Blender

Before (or instead of) exporting, select any mesh and open **Part Inspector** → **Part Stats**. It reports the bounding-box size, the enclosed volume and centre of mass, the surface area, and whether the mesh is *watertight* (a closed manifold) or has open edges — all in the scene's unit. It works on any mesh, not just fitted Reverse objects, and flags the volume as unreliable when the mesh is not closed. Use it to sanity-check a part before committing to an export.

8.2 Verifying a STEP file

- Open it in FreeCAD: it should appear as solid bodies, not a mesh. **Part** workbench → **Check geometry** should report no errors for watertight exports.
- Measure a known dimension (a hole diameter, a plate thickness) — the point of this tool is that those numbers are exact.
- Compare the reported volume in the **Validation Report** against an expectation ($\pi r^2 h$ for a cylinder, etc.).

8.3 Common issues

> **Tip**
>
> Model with intent in Blender and the fits will be exact: keep flat things flat, use enough segments on curved surfaces, apply scale (`Ctrl+A`) before fitting, and prefer one feature per surface.

Symptom	Likely cause / fix
Part imports 1000× too big/small	Unit mismatch. Check the *effective scale* line in the export dialog (Section 7.2).
Holes are filled in the imported part	Exported on the pure-Python path with **Include as solids**, or OCCT not installed. Install OCCT; keep **Cutters** on **Include marked**.
Hole doesn't break through the surface	Increase **Cutter overshoot**, or set the feature's cut mode to **Through**.
Watertight pass reports open edges	Leftover/fillet patches only meet fitted surfaces within the fit deviation. Raise **Sew tolerance** slightly, or exclude leftovers.
Fit has a high RMS	The selection includes faces from another surface. Use the heatmap to spot them, or enable **Reject outliers**.
Auto-detect picks the wrong primitive	Select more of the surface (partial patches are ambiguous), or choose the primitive explicitly — the runner-up list shows how close the call was.
Torus/fillet fit is approximate	Known limitation on partial patches; select as much of the ring/round as possible. Fillet arcs must be $< 330°$.
STEP file is huge	Leftover patches carry many triangles. Remesh, or disable **Include leftover patches**.

Table 3: Troubleshooting quick reference.

9 Roadmap

Development continues along the plan in `PLAN.md` at the repository root: an OCCT-first experience, extruded-profile and revolve recovery (genuine sketch-based solids), fillets and chamfers applied as real blends, multi-body STEP with colours, and — longer term — links from the feature stack into Blender's simulation systems and an optional feature-based CNC toolpath module.

A Running the test suite

From the repository root:

```
# Pure-NumPy core (no Blender needed)
python3 reverse_mesh/tests/test_fitting.py
python3 reverse_mesh/tests/test_step.py
python3 reverse_mesh/tests/test_decompose.py
python3 reverse_mesh/tests/test_solidfit.py

# OCCT kernel round-trip (needs cadquery-ocp)
python3 reverse_mesh/tests/test_occ_export.py

# Blender integration (headless)
blender --background --python reverse_mesh/tests/blender_smoke.py
```

B License

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